Proposal of a novel biomarker for neonatal seizure detection

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Abstract

The abstract serves both as a general introduction to the topic and as a brief, non-technical summary of the main results and their implications. Authors are advised to check the author instructions for the journal they are submitting to for word limits and if structural elements like subheadings, citations, or equations are permitted.

Keywords: Neonatal seizures, Electroencephalography, Power spectral density, Empirical mode decomposition, Neural network, Biomarker

1 Introduction

2 Introduction

Neonatal seizures are the most frequent neurological emergency in newborns. They often signal significant underlying brain dysfunction. The incidence is estimated at 1–5 per 1,000 live births for term neonates. In preterm infants, this rate rises substantially to 57–132 per 1,000 [1]. Seizures are thus more common in the neonatal period than at any other time of life. Common causes include hypoxic–ischaemic encephalopathy, intracranial haemorrhage, ischaemic stroke, and infections of the central nervous system. Inborn errors of metabolism are also a frequent aetiology [2]. Long-term studies link neonatal seizures to adverse outcomes. Affected infants face higher risks of epilepsy, cerebral palsy, cognitive impairment, and behavioural disorders later in childhood [3]. Prognosis depends on rapid identification. Therefore, accurate and timely detection is a critical clinical need.

Electroencephalography (EEG) remains the gold standard for diagnosis [4]. Scalp electrodes capture voltage differences from postsynaptic currents. This provides the necessary resolution to characterise seizure morphology and burden [5]. Continuous monitoring in intensive care has shown that many seizures are subclinical. Estimates suggest 60–80% of electrographic seizures lack clinical signs and escape bedside recognition [6]. Even expert review is not infallible. Inter-observer agreement is often only moderate, with significant discordance reported among clinicians [7]. Manual interpretation is also labour-intensive. Round-the-clock coverage is impractical for many centres. These limitations highlight the urgent need for automated detection systems.

Researchers have explored various automated approaches over the last two decades. Early methods used hand-crafted features from time, frequency, and combined domains. These were classified by traditional algorithms like support vector machines or random forests [8]. Common features included amplitude statistics, spectral powers, and wavelet coefficients [9]. While these systems achieved reasonable sensitivity, they often struggled with high false-detection rates. Generalisability across different recording conditions also proved difficult.

Deep learning has recently gained traction. These architectures offer end-to-end learning. They jointly optimise feature extraction and classification. Convolutional neural networks (CNNs), for instance, learn spatial and spectral representations directly from multichannel EEG [10]. Hybrid models are also promising. Combining CNNs with recurrent layers, such as in a 1D-CNN–LSTM pipeline, has outperformed traditional classifiers on benchmark datasets [11]. In scenarios with limited data, Empirical Mode Decomposition (EMD) can decompose signals into Intrinsic Mode Functions. This enriches the representation and improves sensitivity [12]. Yet, most deep learning models remain opaque. They offer little insight into the physiological signals driving their decisions.

Alongside automated classifiers, spectral characteristics have been investigated as biomarkers. Power spectral density (PSD) analysis quantifies energy distribution across frequencies. It characterizes brain-state changes effectively. During neonatal seizures, the PSD profile often deviates from the typical $1/f^{-\alpha}$ decay seen in rest. Seizure epochs frequently show elevated power in specific bands, such as the theta range (3–12 Hz) [13]. Measures like delta power have been used to predict encephalopathy severity. They can also identify neonates at risk for seizures early in the recording [14].

3 Methods

This study employs two distinct pipelines for feature extraction from neonatal EEG. The first derives compact spectral descriptors. It fits a linear model to the log-transformed power spectral density (PSD) in standard neurophysiological frequency bands. This strategy characterizes a spectrum by its slope, midband-fit value, and intercept. It has been validated in a different biomedical context. Sinha et al. showed that these same three parameters, extracted from photoacoustic signals, could reliably distinguish malignant prostate tissue from benign specimens [15]. Their success suggests these descriptors capture fundamental properties of spectral shape. The second pipeline uses Empirical Mode Decomposition (EMD). This data-driven method decomposes EEG epochs into oscillatory components. Features based on energy and frequency are then computed. Both feature sets undergo dimensionality reduction via Principal Component Analysis. They are then classified by an identical feedforward neural network. This ensures performance differences reflect the feature quality rather than classifier bias.

3.1 Dataset

We analyzed a publicly available neonatal EEG corpus from Helsinki University Hospital. The dataset includes 79 neonates. Seizures were confirmed in 39 infants through consensus annotation. The remaining 40 served as controls. Recordings spanned approximately one hour per subject, with a median of 74 minutes. Three distinct experts annotated seizure events. We retained only epochs with unanimous agreement as ground truth. Epochs with conflicting labels were excluded to minimize noise. Annotations provided onset time and duration at one-second resolution.

We re-referenced the raw recordings to an anterior–posterior bipolar montage. This matched the configuration used during the original annotation. Consistency between the automated pipeline and the human annotators was thus preserved.

3.2 Log-transformed Power Spectral Density and Curve Fitting

Power spectral density (PSD) offers a compact summary of oscillatory content. It quantifies energy distribution across frequencies. We first applied a fourth-order Butterworth bandpass filter. This isolated conventional brain-wave bands: delta (0.7–4 Hz), theta (4–8 Hz), alpha (8–13 Hz), and beta (13–30 Hz). Independent filtering yielded cleaner spectral estimates for each range.

We estimated the PSD for each band and epoch. These were converted to a logarithmic scale. The transform compresses the dynamic range and linearizes the typical $1/f^{-\alpha}$ decay. We then fitted a least-squares regression line to the log-power versus log-frequency curve. This line captures the dominant spectral trend.

Three parameters were extracted from the fitted line: slope, midband-fit, and intercept. The slope quantifies spectral roll-off. The midband-fit represents the predicted log-power at the band's geometric center. The intercept indicates baseline power. This approach follows the framework of Sinha et al. [15]. They found these descriptors effective for tissue classification. We test whether this parameter set can similarly capture the spectral signatures of neonatal seizures. With four bands and three parameters each, the feature vector contained twelve elements.

3.3 Empirical Mode Decomposition Feature Set

Empirical Mode Decomposition (EMD) is a data-driven technique developed by Huang et al. [16] for the analysis of non-linear and non-stationary time series. Unlike Fourier or wavelet-based methods that rely on fixed a priori basis functions, EMD adaptively decomposes a signal $x(t)$ into a finite set of oscillatory components called Intrinsic Mode Functions (IMFs).

The decomposition employs an iterative "sifting" process. For a given signal $r_{i-1}(t)$ (where $r_0(t) = x(t)$), the procedure identifies local maxima and minima, constructing upper and lower envelopes via cubic spline interpolation. The mean of these envelopes, $m_{ik}(t)$, is subtracted from the current proto-mode $h_{ik}(t)$ to refine the component:

$$h_{ik}(t) = h_{i(k-1)}(t) - m_{ik}(t) \tag{1}$$

This sifting iterates until $h_{ik}(t)$ approaches a valid IMF, defined by two conditions: (1) the number of extrema and zero-crossings must be equal or differ by at most one; and (2) the local mean of the upper and lower envelopes is zero at any point. Once an IMF $c_i(t)$ is extracted, it is subtracted from the data to yield a residual $r_i(t) = r_{i-1}(t) - c_i(t)$, and the process repeats until the residual becomes monotonic.

From the extracted IMFs, we computed a set of six statistical and spectral features to capture the morphological characteristics of the seizures:

1. **Standard Deviation (σ):** A measure of the dispersion or amplitude variability of the IMF.
2. **Kurtosis:** A measure of the "tailedness" of the distribution, sensitive to impulsive events:
$$\text{Kurtosis} = \frac{1}{N}\sum_{n=1}^{N}\left(\frac{c_i[n]-\mu}{\sigma}\right)^4 \tag{2}$$
3. **Skewness:** A measure of the asymmetry of the data distribution around its mean:
$$\text{Skewness} = \frac{1}{N}\sum_{n=1}^{N}\left(\frac{c_i[n]-\mu}{\sigma}\right)^3 \tag{3}$$
4. **Energy:** The sum of squared amplitudes, representing the total power within the mode.
5. **Wiener Entropy (Spectral Flatness):** A measure of the flatness of the power spectrum $P(f)$, defined as the ratio of the geometric mean to the arithmetic mean:
$$\text{Wiener Entropy} = \frac{\exp\left(\frac{1}{M}\sum \ln P(f)\right)}{\frac{1}{M}\sum P(f)} \tag{4}$$
6. **PSD RMS:** The root mean square of the power spectral density, summarizing the spectral energy distribution.

These features formed the input to the classification pipeline.

3.4 Discrete Wavelet Transform Feature Set

The Discrete Wavelet Transform (DWT) provides a multi-resolution analysis of time-series data by decomposing a signal into approximation and detail coefficients through successive filtering and downsampling [17]. Unlike EMD, which is entirely data-driven, the DWT uses a fixed mother wavelet as a basis function, offering repeatability and analytical guarantees.

We applied the DWT to each one-second EEG epoch using the Daubechies-4 (`db4`) wavelet at decomposition level 3. This produced four sets of coefficients: one approximation band (A_3) and three detail bands (D_3, D_2, D_1). Each band captures oscillatory content at a progressively finer temporal scale. For a signal $x[n]$, the decomposition proceeds recursively:

$$c_{A,j}[k] = \sum_n h[n-2k]\, c_{A,j-1}[n] \tag{5}$$

$$c_{D,j}[k] = \sum_n g[n-2k]\, c_{A,j-1}[n] \tag{6}$$

where $h[\cdot]$ and $g[\cdot]$ are the low-pass and high-pass filters associated with the `db4` wavelet, and j indexes the decomposition level.

From each coefficient band, we computed the same six statistical features used for the EMD pipeline: energy, Wiener entropy (spectral flatness), skewness, kurtosis, PSD RMS, and standard deviation. With four bands and six features each, the wavelet feature vector contained 24 elements per channel per epoch.

This feature set was processed through the same PCA dimensionality reduction and neural network classification pipeline described in Sections 3.5.

3.5 Evaluation

3.5.1 Dimensionality Reduction and Statistical Testing

We applied Principal Component Analysis (PCA) to both feature sets. This projected them into a comparable low-dimensional space. PCA removed redundant dimensions. It retained the principal variance-bearing components. This ensured the classifier operated on inputs of equivalent complexity.

We also conducted independent-samples Welch t-tests. These assessed the discriminative power of individual components. The resulting t-statistics and p-values offered transparency. They highlighted which dimensions drove group separability.

3.5.2 Classification Architecture

An identical neural network classified both pipelines. This controlled for architectural variables. The network had three fully connected hidden layers with 128, 64, and 32 units. Batch normalization stabilized activations. Rectified Linear Units (ReLU) provided non-linearity. Dropout with a probability of 0.7 mitigated overfitting. The output layer used a softmax activation to produce class probabilities.

Training minimized weighted categorical cross-entropy loss. Weights were inverse-frequency to handle class imbalance. We used the Adam optimizer with a learning rate of 0.001. The batch size was 64. Early stopping monitored validation loss with a patience of 10–25 epochs. Training was capped at 100 epochs.

3.5.3 Post-processing and Threshold Optimisation

A moving-average filter smoothed the raw probabilities. This reduced short-term fluctuations. It suppressed isolated spikes not reflecting sustained activity. We then determined an optimal decision threshold on the validation set. This converted the probability trajectory into a binary classification.

3.5.4 Class-Imbalance Management

Neonatal EEG data is highly imbalanced. Non-seizure epochs vastly outnumber seizure epochs. We used an adaptive nearest-neighbour downsampling strategy. All seizure epochs were kept. Non-seizure epochs were ranked by their temporal distance to the nearest seizure. We selected the closest non-seizure samples based on a target ratio (default 2.0). A minimum count safeguard ensured the non-seizure class remained larger. This approach preserved the temporal context around seizures. This context is crucial for learning discriminative boundaries.

4 Results

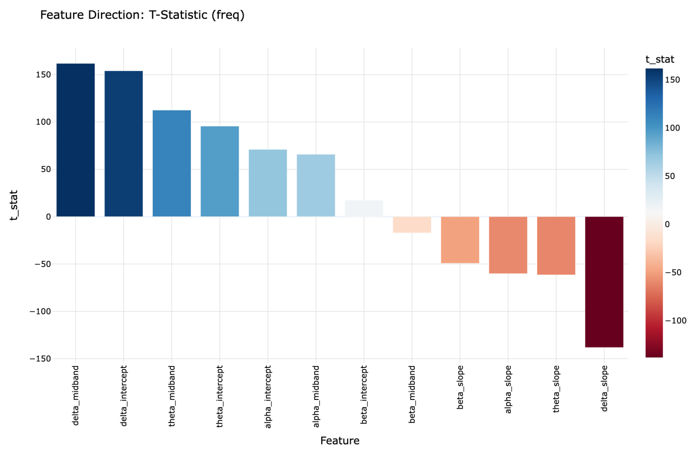
We assessed the discriminative power of individual feature components. Independent-samples t-tests compared seizure and non-seizure groups for each principal component. Figures 1a and 1b show the t-statistics and negative log-transformed p-values for frequency-domain features. Figures 2a and 2b display the underlying statistical results for the EMD features.

Tables 1 and 2 list the top five trials for each pipeline. We ranked them by validation accuracy. Every trial used a unique random seed for data splitting. This allowed us to assess variability across different folds.

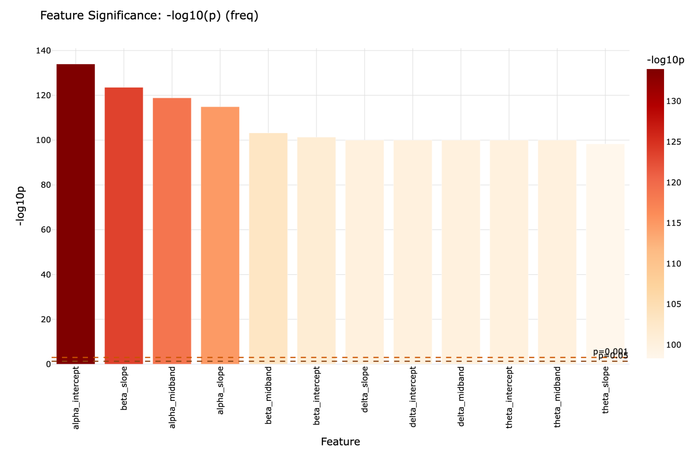
Table 1: Top 5 trials – PCA Frequency Features

	Validation					Test			
Trial	Acc	F1	Prec	Rec	AUC	Acc	F1	Prec	
0	**0.762**	**0.688**	**0.683**	**0.694**	**0.807**	**0.738**	**0.731**	**0.683**	
2	0.721	0.322	0.843	0.199	0.698	0.711	0.480	0.835	
8	0.685	0.554	0.891	0.402	0.767	0.718	0.300	0.864	
5	0.663	0.687	0.599	0.805	0.728	0.411	0.457	0.696	
1	0.639	0.438	0.614	0.340	0.658	0.603	0.503	0.931	
$\mu\pm\sigma$	0.694±0.049	0.538±0.159	0.726±0.134	0.488±0.253	0.732±0.058	0.636±0.136	0.494±0.154	0.802±0.108	0.

The frequency-domain pipeline performed best in Trial 0. It reached a validation accuracy of 0.762. This trial also delivered the strongest test results: 0.738 accuracy, 0.858 AUC, and 0.731 F1 score. Test AUC across the top five trials ranged from 0.464 to 0.858. This indicates moderate sensitivity to the data split. However, discriminative capacity generally remained favourable. Several trials showed consistently high precision. For instance, Trial 8 achieved 0.864 and Trial 1 reached 0.931. High precision sometimes reduced recall. This reflects a conservative tendency in seizure prediction.



(a) Frequency features: t-statistic per principal component



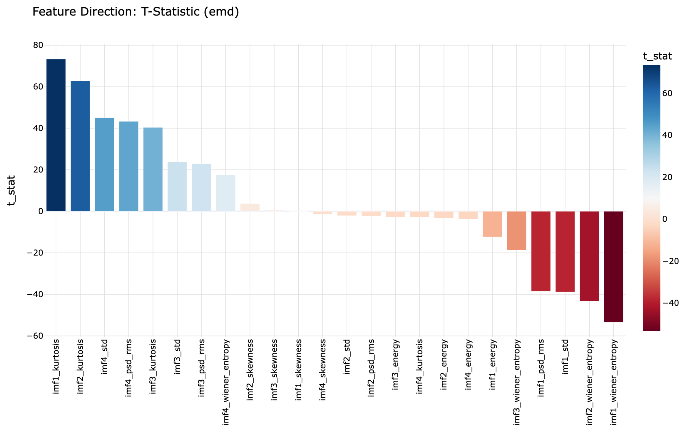
(b) Frequency features: $-\log(p)$ per principal component

Fig. 1: Statistical comparison of PCA-reduced frequency-domain features between seizure and non-seizure epochs.

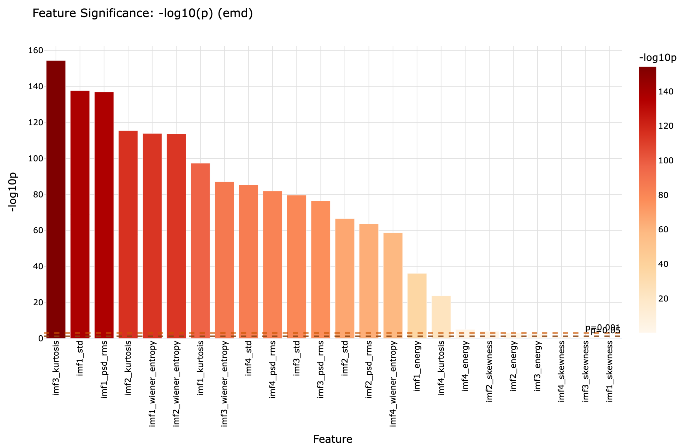
The EMD pipeline showed lower overall accuracy. Its best validation score was 0.672 (Trial 0). The corresponding test AUC was 0.779. However, EMD features often produced higher recall. Trial 0, for example, achieved a recall of 0.843. This suggests greater sensitivity to seizure events, though precision was lower. Cross-trial variability was more pronounced here. Test AUC values spanned from 0.508 to 0.779.

To examine the performance landscape further, we generated radar charts. Figure 3 compares validation metrics for the top performing trials of both pipelines.

These charts align with the tabular findings. The frequency-domain features provide a robust all-around classifier, evidenced by the larger polygon area. EMD features



(a) EMD features: t-statistic per principal component



(b) EMD features: $-\log(p)$ per principal component

Fig. 2: Statistical comparison of PCA-reduced EMD features between seizure and non-seizure epochs.

capture complementary information. They prioritise sensitivity. This performance difference is structural. It suggests EMD-derived metrics may be valuable in screening. In such clinical contexts, missing a seizure is far worse than a false alarm.

4.1 Wavelet Feature Results

We also evaluated the DWT-based feature pipeline. Figures 4a and 4b present the t-statistics and significance levels for the PCA-reduced wavelet features.

Table 2: Top 5 trials – PCA EMD Features

	Validation					Test			
Trial	Acc	F1	Prec	Rec	AUC	Acc	F1	Prec	Rec
0	**0.672**	**0.647**	**0.546**	**0.792**	**0.790**	**0.678**	**0.703**	**0.603**	**0.843**
5	0.655	0.656	0.605	0.717	0.712	0.388	0.392	0.711	0.270
3	0.652	0.741	0.650	0.861	0.660	0.590	0.541	0.432	0.724
8	0.646	0.638	0.636	0.639	0.674	0.602	0.497	0.430	0.588
4	0.632	0.583	0.668	0.517	0.698	0.604	0.652	0.576	0.753
$\mu\pm\sigma$	0.651±0.015	0.653±0.057	0.621±0.048	0.705±0.134	0.707±0.051	0.572±0.109	0.557±0.124	0.550±0.120	0.636±0.

Table 3 lists the top five trials for the wavelet pipeline, ranked by validation accuracy.

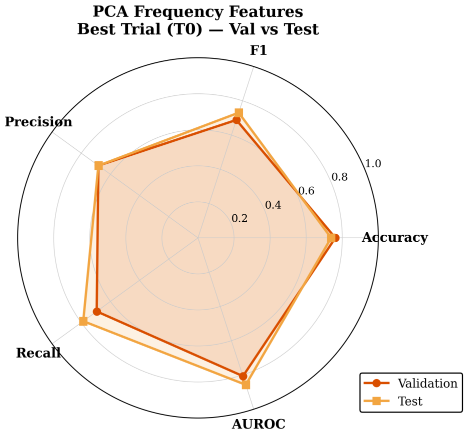
Table 3: Top 5 trials – PCA Adaptive Wavelet Features

	Validation					Test			
Trial	Acc	F1	Prec	Rec	AUC	Acc	F1	Prec	Rec
5	**0.681**	**0.566**	**0.635**	**0.511**	**0.703**	0.434	0.339	0.756	0.219
6	0.649	0.756	0.708	0.811	0.585	0.420	0.568	0.399	0.985
4	0.623	0.485	0.603	0.406	0.671	0.651	0.614	0.625	0.604
3	0.609	0.566	0.617	0.524	0.654	**0.681**	**0.564**	**0.517**	**0.620**
1	0.582	0.564	0.472	0.702	0.645	0.621	0.607	0.593	0.620
$\mu\pm\sigma$	0.629±0.038	0.587±0.100	0.607±0.086	0.591±0.163	0.652±0.043	0.561±0.125	0.538±0.114	0.578±0.132	0.610±0.

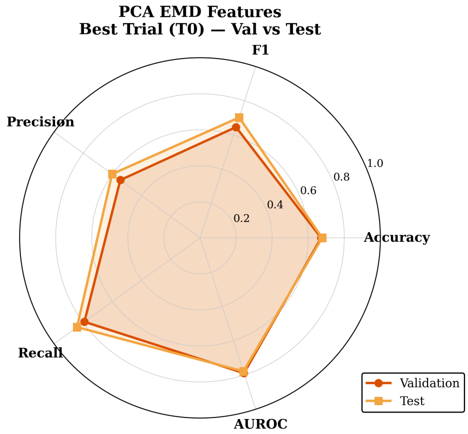
The wavelet pipeline yielded moderate validation accuracy, peaking at 0.681 in Trial 5. Notably, validation-to-test generalisation varied considerably across trials. Trial 5, despite ranking first in validation accuracy, dropped to 0.434 on the test set. In contrast, Trial 3 demonstrated the strongest test performance with 0.681 accuracy and 0.727 AUC. Trial 4 also generalised well, achieving a test AUC of 0.697 and an F1 score of 0.614. Trial 6 showed extreme recall (0.985) on the test set but at the expense of precision (0.399), reflecting an aggressive detection strategy.

Compared to the frequency-domain and EMD pipelines, the wavelet features exhibited a wider generalisation gap. This may reflect the higher-dimensional raw feature space (24 elements versus 12 for frequency), which—even after PCA—can introduce greater sensitivity to partition-specific structure. Nevertheless, the wavelet pipeline's best test AUC (0.727) remained competitive. The multi-resolution decomposition captured complementary spectral dynamics not accessible through global PSD fitting or adaptive EMD.

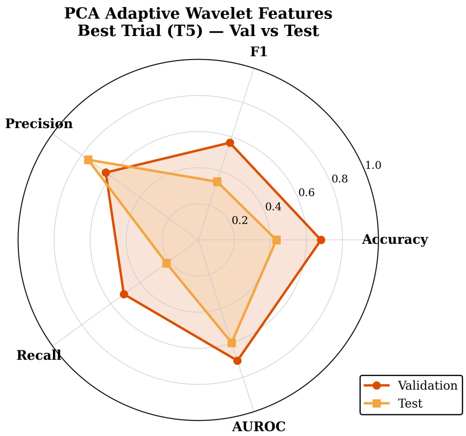
Discussions should be brief and focused. In some disciplines use of Discussion or 'Conclusion' is interchangeable. It is not mandatory to use both. Some journals prefer a section 'Results and Discussion' followed by a section 'Conclusion'. Please refer to Journal-level guidance for any specific requirements.



(a) Frequency Features



(b) EMD Features

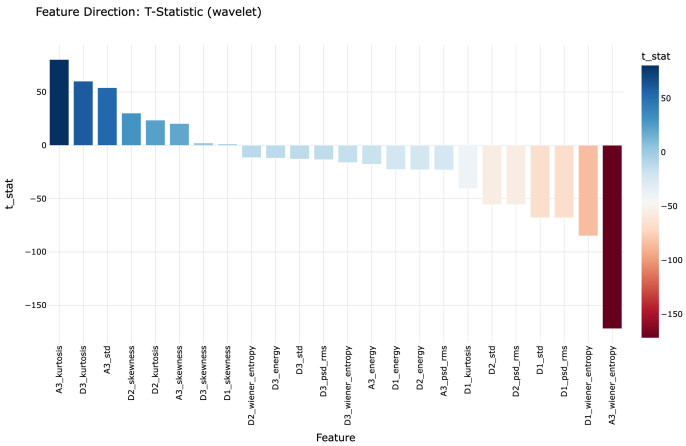


(c) Wavelet Features

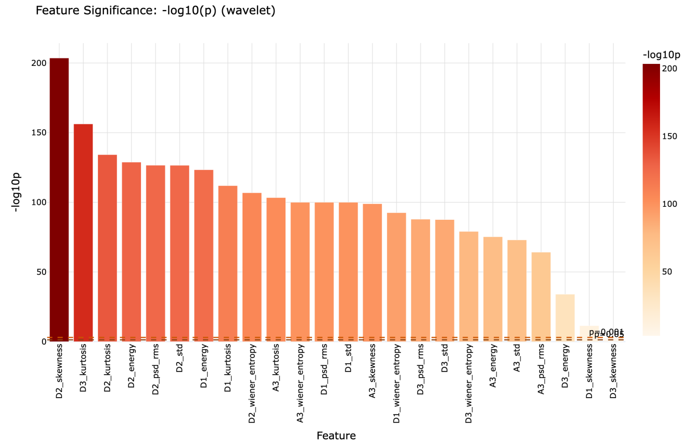
Fig. 3: Radar charts illustrating the performance profile of the top trials. The frequency-domain pipeline (a) demonstrates a more balanced profile with high Area Under the Curve (AUC) and Accuracy. The EMD pipeline (b) exhibits a distinct skew towards Recall, indicating superior sensitivity to seizure events despite lower overall precision. The Wavelet pipeline (c) presents an intermediate profile, capturing complementary spectral features.

5 Conclusion

Conclusions may be used to restate your hypothesis or research question, restate your major findings, explain the relevance and the added value of your work, highlight any limitations of your study, describe future directions for research and recommendations.



(a) Wavelet features: t-statistic per principal component



(b) Wavelet features: $-\log(p)$ per principal component

Fig. 4: Statistical comparison of PCA-reduced wavelet features between seizure and non-seizure epochs.

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Supplementary information. Not applicable.

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Declarations

- **Funding:** Not applicable.
- **Conflict of interest:** The authors declare no competing interests.
- **Ethics approval:** Not applicable. This study used a publicly available, de-identified dataset.
- **Data availability:** The neonatal EEG dataset used in this study is publicly available from the Zenodo repository associated with Stevenson et al. (2019).
- **Code availability:** Not applicable.
- **Author contribution:** Not applicable.

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